Rectification of Harmonic Embeddings, LNDs and the Makar-Limanov Invariant

Research Appendix: Results, Exact Certificates, and Open Problems

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Abstract

We consolidate the results of a research branch starting from explicit complex rectifications of the Shastri trefoil and a harmonic figure-eight. Consequences for specific conjectures of van den Essen–van Rossum are distinguished from the general embedding, cancellation, and linearization problems, which are not resolved here. We give constructive passages to locally nilpotent derivations, exact minimum slice degrees, formal kernel coordinates, and filtration formulas for classes of affine modifications. Rectifiability of an embedded line is characterized by the existence of a normal locally finite vector field, and finite certificates for the search for such fields are provided. For $H(3,7,8)$ and $H(4,5,7)$, complete affine spaces of candidates of dimensions 24 and 19 are obtained; local finiteness and new complex rectifications have not been established. The appendix includes reproducible computations and precise limits on every assertion. No motivic methods, K-theory, or homological algebra are used.

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Research branch. All material presented in this appendix remains under verification. This branch continues the study of consequences of the known rectifications and their connections with the Abhyankar–Moh embedding problem, locally nilpotent derivations, the Makar-Limanov invariant, and affine geometry. The authors consider it useful to share these interim results, proofs, and computations before verification is complete. They hope that this direction will develop into a broader research programme and invite all interested readers to check, discuss, and participate in the investigation.

Provenance and status. This material was prepared through interactive work with AI; individual arguments and implementations were checked within the same process. This is not independent external peer review. No claim of priority is made. Below, “proved” means that a mathematical proof and, where applicable, an exact certificate are supplied; it does not mean that verification by the research community is complete.

Source: the abhyankar-moh-trefoil repository, verified commit `b954f3609e900`. This text is a separate continuation of the appendix dated 26 October 2026 and the full English counterpart of the Russian research text dated 28 October 2026.

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1 Register of resolved and unresolved questions

Five statuses must be distinguished in this work: an original certified result of the repository; a proved consequence of that result; a new assertion of the working appendix accompanied by a proof; a finite computational result; and an open question. Failure of a bounded search is not regarded as a proof of nonrectifiability. Several equivalent formulations concerning a single candidate are not counted as solutions of several general problems.

Question	Precise outcome	Scope of the conclusion
Shastri trefoil	Mutually inverse rectifying maps are supplied over $K = \mathbb{Q}(\sqrt{-3})$.	Nonrectifiability is refuted for this specific complex model; the real knot remains nontrivial.
Harmonic figure-eight	$H(3,5,7)$ is rectified, with linear term $+5u$ in its second component.	This model differs from Shastri's Example 2: there the middle component is $u^5 - 5u^3 + 4u$.
Conjecture 5.1 in [6]	For the specified D, its kernel is isomorphic to $\mathbb{C}^{[4]}$, contrary to the conjecture.	This particular candidate is not a counterexample to cancellation.
Conjecture 5.2 in [6]	The original slice is a coordinate; its zero fibre is $\mathbb{A}^4_{\mathbb{C}}$.	A consequence of polynomiality of the same kernel.
Conjecture 5.3 in [6]	The associated extended Rees algebra is polynomial.	The paper calls this particular statement Asanuma's Conjecture; other conjectures of Asanuma are not addressed.
Harmonic families	All admissible $H(3,4,c)$ and $H(3,5,c)$ are rectifiable over K.	This is neither a classification of all $H(a,b,c)$ nor a claim of infinitely many new real knot types.
Slice degrees	A complete reduction to finite linear algebra, eight exact minima, and the full dependence on multiplicity for two models.	Degree depends on the original coordinates and is not a conjugacy invariant.
Normal field	A rectification criterion through local finiteness is proved, together with a constructive recovery of coordinates.	Existence of such a field for an arbitrary embedding has not been proved.
$H(3,7,8)$ and $H(4,5,7)$	Spaces of candidate fields, degree obstructions and several excluded classes are certified.	Complex rectifiability, polynomiality of the new kernels and their full ML are not established here.

Question	Precise outcome	Scope of the conclusion
General problems	The spatial embedding problem, all harmonic triples, cancellation and general linearization are not solved here.	Saying “not solved by this work” does not replace a separate survey of all known versions of these problems.

Historical accuracy. The planar Abhyankar–Moh–Suzuki theorem is already a classical result [2, 3]. The historical conjecture that nonrectifiable space lines exist is not refuted by rectifying one or two candidates. Questions 1 and 2 in Shastri’s paper [4], concerning the existence of polynomial representations of knots that extend to complex embeddings, were solved by Shastri himself. Our conclusion concerns the specific conjecture about his trefoil recorded in [6, p. 1319].

Results not claimed. We have not proved the general Abhyankar–Sathaye conjecture, tameness or wildness of the constructed automorphisms, minimality of the degrees of the rectifying maps themselves, a general lower bound of $6n+1$ for slice degrees, or the exact general factorization complexity of successive modifications. These questions remain part of the research programme and are not included among the resolved problems.

2 Original rectifications and harmonic families

2.1 Notation and the verified original construction

The field k has characteristic zero. We normalize the Chebyshev polynomials by

$$C_j(u) = 2T_j(u/2), \qquad C_0 = 2, \quad C_1 = u, \quad C_{j+1} = uC_j - C_{j-1}.$$

We consider closed embeddings, that is, $k[f_1, f_2, f_3] = k[u]$. A parameter polynomial P satisfies $P \circ \gamma = u$; to be a coordinate, it must additionally extend to a polynomial automorphism. These two conditions are not equivalent.

The original preprint [1] provides automorphisms Φ_T, Φ_H and their polynomial inverses over $K = \mathbb{Q}[a]/(a^2 - a + 1) \simeq \mathbb{Q}(\sqrt{-3})$ for

$$\gamma_T(u) = (u^3 - 3u, u^4 - 4u^2, u^5 - 10u),$$
$$\gamma_H(u) = (C_3(u), C_5(u), C_7(u)).$$

The identities $\Phi_T\gamma_T = \Phi_H\gamma_H = (u, 0, 0)$, both inverse compositions, and all twelve printed components have been verified by exact polynomial identities. The component degrees are respectively $(5, 15, 24)$ and $(7, 21, 41)$; their minimality is not asserted. The complete formulas are retained in the original preprint and in the self-contained computational package. The change of coordinates

$$M(x, y, z) = (x, y + 2, z - 5x)$$

takes γ_T to (C_3, C_4, C_5).

2.2 Constructive transfer through a modification

Let $A = (x, H, W)$ and $A^{-1} = (x, \mathcal{Y}, \mathcal{Z})$ be mutually inverse automorphisms, with $m \in k[x]$, $\deg m > 0$, $p \in k[x, y]$. Consider $\beta = (x, p(x, y) + m(x)z, y)$. In the quotient ring $S = k[x]/(m)$, set $c(y) = H(x, p(x, y), y)$. If the linear coefficient of c is a unit and all higher coefficients are nilpotent, the substitution $y \mapsto c(y)$ defines an automorphism of $S[y]$. Its compositional inverse ψ is obtained by a finite lifting process along the nilpotent ideal generated by the higher coefficients.

Choose a polynomial representative

$$\varphi(x, T) \equiv W(x, p(x, \psi(T)), \psi(T)) \pmod{m}.$$

Then

$$\mathcal{A} = \left(x, H \circ \beta, \frac{W \circ \beta - \varphi(x, H \circ \beta)}{m} \right)$$

and

$$\mathcal{A}^{-1}(x, T, Q) = \left(x, \mathcal{Z}(x, T, mQ + \varphi), \frac{\mathcal{Y}(x, T, mQ + \varphi) - p(x, \mathcal{Z}(x, T, mQ + \varphi))}{m} \right)$$

have polynomial components: both numerators are divisible by the full m, including the multiplicities of its roots. After inverting m, both compositions are the identity. Injectivity of localization carries these identities back to the original ring, including the exceptional fibre $m = 0$. This explicitly implements classical lifting and relative-coordinate mechanisms; the general transfer idea is not claimed as new [8, 1]. The case $m \in k^*$ is simpler: β itself is already an automorphism.

If $H \circ \gamma = u$ and $w = W \circ \gamma$, the relative pair yields a complete rectifying map

$$\Phi = (H,\ x - f_1(H),\ W - w(H)).$$

A parameter polynomial alone, without a verified coordinate pair, does not suffice for this conclusion.

2.3 Polynomial changes of harmonic indices

For coprime a, b and signs $\varepsilon_a, \varepsilon_b \in \{\pm 1\}$, the congruences

$$c' \equiv \varepsilon_a c \pmod{2a}, \qquad c' \equiv \varepsilon_b c \pmod{2b}$$

imply

$$C_{c'} - \varepsilon_a \varepsilon_b C_c \in \mathbb{Q}[C_a, C_b].$$

Consequently, the corresponding space curves are related by a polynomial shear of the third coordinate.

A short proof uses the nodes of the plane projection (C_a, C_b). Its coordinate ring consists exactly of the polynomials in u that have equal values at the two preimages of every node: inclusion is immediate, and equality follows by comparing codimensions $(a-1)(b-1)/2$ through the semigroup $\langle a, b \rangle$ and interpolation. The stated congruences of indices give the required equalities of values by the trigonometric Chebyshev formula. Eliminating leading terms produces a shear polynomial with rational coefficients. Details and exact examples are given in the original appendix; the nodal projection is treated in [9].

This gives two complete families over K:

$$H(3,4,c), \quad \gcd(c,12)=1; \qquad H(3,5,c), \quad \gcd(c,15)=1.$$

For the pair $(3,4)$, the representatives $c=1,5$ modulo 24 suffice; for $(3,5)$, they are $c=1,2,4,7$ modulo 30. Representatives with index 1 are immediate, those with index 2 are classically known, and $c=4$ reduces by a permutation to $(3,4,5)$. The remaining two are supplied by the original rectifications.

Index 1, the admissible models $H(2,b,c)$ and $H(a,b,a+b)$ were already known in the literature [9]; they are not attributed to this work. The topological classification of harmonic knots [10] is likewise not a classification of complex embeddings up to automorphisms.

2.4 Finite enumeration and its precise scope

All 44142 pairwise coprime triples $2 \le a < b < c \le 100$ have been enumerated using every strictly decreasing elementary reduction

$$c = ra + sb \quad \longmapsto \quad |ra - sb|, \qquad (x,y,z) \longmapsto (x,y,C_r(x)C_s(y) - z).$$

Of these, 3209 reach index 1, another 237 reach a verified base model, and 40696 remain unresolved *by this algorithm*. The last number is not a count of nonrectifiable curves.

For any embedding $\gamma = (f_1, \dots, f_n)$, we subsequently use

$$A = k[t,u,X_1,\dots,X_n], \qquad D_\mu = t^\mu \partial_u + \sum_i f_i'(u)\partial_{X_i}, \qquad V = \gamma(u) - t^\mu X,$$

where $\mu \ge 1$. This is an LND in $n+2$ variables. It must not be confused with the Jacobian LND of a modification in three variables.

3 The minimum slice degree theorem

Set $V = \gamma(u) - t^\mu X$. For every $P \circ \gamma = u$, there is a polynomial slice

$$S_{P,\mu} = \frac{u - P(V)}{t^\mu}, \qquad D_\mu(S_{P,\mu}) = 1.$$

Polynomiality follows from exact divisibility of the numerator; the displayed denominator does not make this a nonpolynomial rational function. This classical construction goes back to van den Essen–van Rossum [6]. The calculations below refine the degree-minimization problem.

Theorem 3.1. Completeness of the reduction

The minimum degree among all polynomial slices equals the minimum of $\deg S_{P,\mu}$ over all parameter polynomials P.

Proof. Introduce the weights $\mathrm{wt}(t)=1$, $\mathrm{wt}(u)=0$, $\mathrm{wt}(X_i)=-\mu$. The derivation raises weight by μ. If $D_\mu(s)=1$, the component $s_{-\mu}$ of weight $-\mu$ is also a slice; its ordinary degree is no greater than that of s. After inverting t,

$$A_t = k[t,t^{-1},u,V], \qquad D_\mu = t^\mu \partial_u.$$

The element $t^{\mu}s_{-\mu} - u$ is invariant and has weight zero. Since all V_i have weight zero, it equals $-P(V)$ for a unique $P \in k[V]$. Reducing the exact polynomial identity modulo t gives $P \circ \gamma = u$. □

Theorem 3.2. Exact formula and finite criterion

With the convention $\deg 0 = -\infty$, we have

$$\boxed{\deg S_{P,\mu} = \max_{\alpha \neq 0}\{\deg_u((\partial^{\alpha}P) \circ \gamma) + (\mu+1)|\alpha| - \mu\}.}$$

Indeed, Taylor's formula gives

$$S_{P,\mu} = \sum_{\alpha \neq 0} \frac{(-1)^{|\alpha|+1}}{\alpha!} t^{\mu(|\alpha|-1)} X^{\alpha} (\partial^{\alpha}P)(\gamma(u)).$$

Distinct α correspond to distinct monomials in X, so they cannot cancel one another. A nonzero derivative of order $\deg P = r$ gives the lower bound $(\mu+1)r - \mu$.

Consequently, a slice of degree $\leq d$ exists if and only if the following finite linear system is solvable:

$$\deg P \leq \left\lfloor \frac{d+\mu}{\mu+1} \right\rfloor, \qquad P \circ \gamma = u,$$
$$[u^j](\partial^{\alpha}P)(\gamma(u)) = 0 \quad \text{for } j > d - (\mu+1)|\alpha| + \mu, \quad \alpha \neq 0.$$

It is linear in the coefficients of P. For $n = 3$, the number of unknowns is at most $\binom{\lfloor (d+\mu)/(\mu+1) \rfloor + 3}{3}$. This is a complete criterion, rather than a search restricted to a chosen form of slice.

Theorem 3.3. All sufficiently large multiplicities

Let

$$M = \max_i \deg f_i, \qquad r_{\gamma} = \min\{\deg P : P \circ \gamma = u\}, \qquad m_{\gamma}(\mu) = \min_{D_{\mu}(s)=1} \deg s.$$

Then

$$\boxed{m_{\gamma}(\mu) = (\mu+1)r_{\gamma} - \mu \quad \text{for } \mu \geq M - 1.}$$

For a nonconstant P of degree r, a derivative of order q has degree at most $M(r-q)$ after substitution. When $\mu + 1 \geq M$,

$$M(r-q) + (\mu+1)q - \mu \leq (\mu+1)r - \mu.$$

A nonzero derivative of order r attains this bound. Minimizing over P completes the proof. Moreover, $m_{\gamma}(\mu)$ is nondecreasing: every expression in the maximum has nonnegative slope $|\alpha| - 1$.

This degree measure depends on the original coordinates. It is not an invariant of conjugacy of LNDs and, without a further argument, cannot serve as an obstruction to polynomiality of the kernel.

4 Exact calculations for harmonic models

For $\mu = 1$, the following results were obtained. The lower bounds are certified by rational dual certificates, and the upper bounds by explicit slices.

Model	$m_\gamma(1)$	Unknowns	Nonzero weights
$H(3,4,5)$	7	20	11
$H(3,5,7)$	11	56	35
$H(4,5,7)$	12	84	41
$H(3,7,8)$	13	84	41
$H(3,10,11)$	19	220	111
$H(3,13,14)$	25	455	225
$H(5,7,8)$	11	56	29
$H(7,10,11)$	15	120	61

The last two columns refer to the system excluding degree one below the minimum. The first two rows reproduce earlier results with more compact certificates.

If the system is written as $Ac = b$, a lower-bound certificate is a rational vector λ satisfying $\lambda^{\mathsf{T}} A = 0$ and $\lambda^{\mathsf{T}} b = 1$. This rules out a solution over every field of characteristic zero, not only over $\mathbb{Q}$. A separate verification script rebuilds the columns and performs the calculations using `Fraction`; no symbolic solver is needed for verification.

$H(3,7,8)$: a lower-degree retraction is not always better

The Chebyshev identities give

$$C_7 C_8 = C_{15} + C_1, \qquad C_5(C_3) = C_{15}, \qquad P_5 = yz - C_5(x) = yz - x^5 + 5x^3 - 5x.$$

The resulting slice of degree 13 is optimal for $\mu = 1$. However, the minimum degree of the parameter polynomial itself is 4; one may take

$$\begin{aligned} P_4 = {} & x^3 z + x^3 - x^2 yz - xy^2 - 2xz - x \\ & + y^3 + 2yz - 2y. \end{aligned}$$

For $\mu = 1$, this P_4 yields a slice of degree 19. Thus minimizing $\deg P$ and minimizing $\deg S_P$ are different problems.

The full dependence on multiplicity is

$$\boxed{\begin{aligned} & m_{378}(1) = m_{378}(2) = 13, \quad m_{378}(3) = 17, \quad m_{378}(4) = 21, \\ & m_{378}(5) = 22, \quad m_{378}(6) = 23, \quad m_{378}(\mu) = 3\mu + 4 \quad (\mu \geq 7). \end{aligned}}$$

For $\mu \leq 4$, P_5 suffices; from $\mu = 5$ onward, P_4 suffices. The lower bounds for $1 \leq \mu \leq 7$ have been checked exactly; the last test also excludes $\deg P \leq 3$. Theorem 3.3 applies thereafter.

$H(4,5,7)$: a formula for all multiplicities

Here

$$P_3 = (x^2-2)z - y^3 + 3y, \qquad \boxed{m_{457}(\mu) = \max\{12, \mu+9, 2\mu+3\}.}$$

The first, second, and third derivatives of P_3 along the curve have maximum degrees 11, 7, and 0. This gives the upper bound. For $\mu = 1,2,3$, the lower bound 12 follows from the certificate at $\mu = 1$ and monotonicity.

Restriction to the curve is injective on polynomials of degree ≤ 3: the corresponding 20×20 minor has determinant 267386. Therefore P_3 is unique in this degree range. For $\mu = 4,5$, every P of degree ≥ 4 gives a slice of degree at least $3\mu+4 > \mu+9$. For $\mu \ge 6$, the general theorem applies with $M = 7$, $r_\gamma = 3$.

An infinite family of upper bounds

For odd $a, m \ge 3$, set $b = (ma-1)/2$, $c = (ma+1)/2$. The indices are pairwise coprime, and $P = yz - C_m(x)$ recovers the parameter. The exact degree of its slice is

$$\deg S_{P,\mu} = \max\{a(m-1)+1, (\mu+1)m - \mu\}.$$

For $(a,b,c) = (3, 3n+1, 3n+2)$, this gives $m_\gamma(1) \le 6n+1$. Equality is certified for $n = 1,2,3,4$; the general lower bound has not yet been proved.

5 The kernel and formal triviality

Let $I = \ker(k[Y_1, \dots, Y_n] \to k[u])$ be the ideal of the embedding, and let $B_\mu = \ker D_\mu$. Taylor expansion yields the following description, classical in nature:

$$\boxed{B_\mu = k[t, V, I(V)/t^\mu] \simeq k[Y][t, I/t^\mu].}$$

For $\mu = 1$, the corresponding construction is described in [6, Proposition 5.4, Corollary 5.5]. It does not assert polynomiality of B_μ.

The elementary key is this: if all derivatives of F of orders $< q$ vanish along γ, then $F \in I^q$. To prove this, choose $P \circ \gamma = u$ and put $g_i = Y_i - f_i(P(Y))$. Taylor expansion about $\gamma(P(Y))$ first shows that $I = (g_1, \dots, g_n)$, and then gives the required membership. The converse follows from the Leibniz rule. For $j < 0$, a weight component $t^j F(V)$ is polynomial exactly when $F \in I^{\lceil -j/\mu \rceil}$; this gives the kernel formula.

Theorem 5.1. All finite thickenings are trivial

For every embedding, every $\mu \ge 1$, and every $N \ge 1$,

$$\boxed{B_\mu/(t^N) \simeq (k[t]/(t^N))^{[n]}.}$$

The isomorphisms can be chosen compatibly in N, so

$$\boxed{\widehat{B}_\mu^{(t)} \simeq k[p, q_1, \dots, q_{n-1}][[t]].}$$

The right-hand side consists of power series in t with polynomial coefficients. The degrees of these coefficients in p, q may grow. This is not a claim about the ordinary polynomial ring $k[[t]][p,q]$.

Proof. Set $a(u) = \nabla P(\gamma(u))$, $v(u) = \gamma'(u)$. The chain rule gives $av = 1$. Over the PID $k[u]$, the module of rows annihilating v is free of rank $n-1$; choose a basis $\ell_1, \dots, \ell_{n-1}$. Together with a, these rows form an invertible matrix over $k[u]$.

A row ℓ can be lifted explicitly to the ideal of the curve:

$$F_\ell(Y) = \sum_i \ell_i(P(Y))(Y_i - f_i(P(Y))).$$

We have $F_\ell \in I$ and $\nabla F_\ell(\gamma) = \ell - (\ell v)a = \ell$. Define

$$p = P(V), \qquad q_j = -F_{\ell_j}(V)/t^\mu, \qquad s = (u - P(V))/t^\mu.$$

These are polynomials; p, q_j are invariant, and $D_\mu(s) = 1$. At $t = 0$, the change $(u, X) \mapsto (p, q, s)$ is

$$(u, X) \longmapsto (u, \ell_1(u)X, \dots, \ell_{n-1}(u)X, a(u)X)$$

and is invertible.

After composing with its polynomial inverse at $t = 0$, we obtain an endomorphism $\Phi \equiv \mathrm{id} \pmod{t^\mu}$. On $A/(t^N)$, the operator $E = \mathrm{id} - \Phi$ raises t-adic order by μ, so $E^L = 0$ for $L = \lceil N/\mu \rceil$. The inverse map is the finite sum $\sum_{j=0}^{L-1} E^j$. It is the inverse of a ring homomorphism as an additive map and is therefore automatically multiplicative.

Thus

$$A/(t^N) = (k[t]/(t^N))[p, q_1, \dots, q_{n-1}, s], \qquad \overline{D}_\mu = \partial_s.$$

The global slice theorem gives $A = B_\mu[s]$. Consequently, the kernel of the reduced derivation is $B_\mu/(t^N)$. The same polynomials p, q work for every N, ensuring compatibility. □

Consequence for the strategy. The abstract type of any one finite thickening of the fiber $t = 0$ cannot distinguish these kernels from a polynomial kernel. One must control global algebraization, degrees of the inverse map, or additional structures. Finite certificates for degree restrictions remain useful: they retain the original coordinates, which are absent from the abstract thickening ring.

6 A restriction on coordinates for the cubic family

Let $\gamma_n = (C_3, C_{3n+1}, C_{3n+2})$, $n \geq 1$. Over $\mathbb{C}$, there is no polynomial coordinate H satisfying $H \circ \gamma_n = u$ of either form

$$\boxed{H = Q(x, y) + M(x)z \quad \text{or} \quad H = Q(x, z) + M(x)y.}$$

The degrees of Q and M are unrestricted. The exclusion also applies to an absolute coordinate: the full automorphism is not required to preserve x.

Proof of the relative statement

Write $U_j = U_j(x/2)$ for the Chebyshev polynomials of the second kind. In the free module with basis $1, u, u^2 - 2$ over $\mathbb{C}[x]$, where $u^3 - 3u = x$, we have

$$y = U_n u - U_{n-1}(u^2 - 2), \qquad z = -U_{n-1}u + U_n(u^2 - 2).$$

For the first form, reduce Q modulo the monic relation $y^3 - 3y - C_{3n+1}(x) = 0$ to $Ay^2 + By + C$. Comparing coefficients gives

$$U_{3n}A + U_{2n}M = U_{n-1}. \tag{6.1}$$

When $M = 0$, this equality is impossible by degree. If (x, H, W) is an automorphism and $M(\alpha) = 0$, the polynomial $Q(\alpha, y)$ must be affine-linear in y. Hence $A(\alpha) = 0$ and $U_{n-1}(\alpha) = 0$.

All roots of M belong to the real roots of U_{n-1}. The two largest roots of U_{3n}, $x_j = 2\cos(j\pi/(3n+1))$, $j = 1, 2$, lie to the right of all these roots. For $n = 1$, U_0 has no roots at all. Substitution in (6.1), together with the sine formula, gives $M(x_1) = 1$, $M(x_2) = -1$. But for $M = c\prod(x-\alpha)^{e_\alpha}$, the ratio $M(x_1)/M(x_2)$ is positive, even when c is complex. This is a contradiction.

For the second form, one similarly obtains

$$U_{3n+1}A + U_{2n}M = U_n.$$

The two largest roots of U_{3n+1} lie to the right of the roots of U_n and again force opposite values of M. This completes the relative proof.

Passage to an absolute coordinate

For $M \neq 0$, the polynomial H is a coordinate over $\mathbb{C}(x)$. If it were an absolute coordinate, Proposition 3.1 of Edo–van den Essen–Maubach [8] would make it a coordinate over $\mathbb{C}[x]$. This contradicts what has just been proved.

For $n = 1$, this exclusion is compatible with the project's complex constructions for $H(3, 4, 5)$: the parameter component must depend nonlinearly on both transverse variables. This restricts the form of a rectifier; it does not prove that no rectifier exists.

The natural modification also fails

Consider

$$\beta_n = (x, C_n(x)z - y^2 + 2, y), \qquad \beta_n \circ \gamma_n = (C_3, C_2, C_{3n+1}).$$

At every root α of C_n, the center $H(\alpha, 2-y^2, y)$ cannot be affine-linear in y if H recovers the parameter of the target triple. Otherwise, in the quotient by $u^3-3u-\alpha$, we obtain

$$u = \lambda(U_n(\alpha/2)u - U_{n-1}(\alpha/2)(u^2 - 2)) + \nu.$$

Here $U_{n-1}(\alpha/2) \neq 0$, because $C_n(x)^2 + (4 - x^2)U_{n-1}(x/2)^2 = 4$. Comparing coefficients first gives $\lambda = 0$ and then a contradiction. This proof imposes no bound on the degree of H.

For odd n, the target triple is not an embedding: $u = \pm\sqrt{3}$ have the same image. For even $n = 2m$, the target has an explicit relative coordinate. If $C_{6m+1}(u) = uP_m(u^2 - 2)$, $\varepsilon = P_m(1) = (-1)^m$, then

$$A_m(Y) = \frac{1 - \varepsilon P_m(Y)}{Y - 1}, \qquad H_0 = xA_m(Y) + \varepsilon Z$$

recovers the parameter. Thus even rectifiability of the target does not repair the bad center of this particular transfer.

An additional finite result for $H(3,7,8)$

Coordinates $Q(x,L)+M(x)N$ are excluded when (L,N) is any invertible constant linear change of (y,z), $\deg M \le 10$, and the degree of Q is arbitrary. After normalizing $L = y + az$, the problem reduces to 56 exact conditions on the multiplicities of the roots of M. In every case, the gcd of the necessary polynomial conditions is 1; repeated roots at $a = \pm i/2$ are checked separately. This is a **bounded result**: the case $\deg M > 10$ is not excluded here.

7 H(4,5,7): the degree-four barrier in the relative class

Suppose that $F \circ (C_4, C_5, C_7) = u$ and $\deg F \le 4$. Then:

- the unique such F that is a coordinate of $k(x)[y,z]$ is $F_0 = (x^2-2)z - y^3 + 3y$;
- no such F is a coordinate of $k(y)[x,z]$ or $k(z)[x,y]$;
- F_0 is not a coordinate over $k[x]$: its derivatives with respect to y, z vanish simultaneously at $x^2 = 2$, $y^2 = 1$.

Consequently, any method that completes one of the pairs (x,F), (y,F), or (z,F) to an automorphism necessarily requires $\deg F \ge 5$.

The proof exhausts the entire space of polynomials of degree ≤ 4. The restriction matrix for the 35 monomials has rank 28. All parameter polynomials form the affine space $F_0 + \langle K_0, \dots, K_6 \rangle$. The archive lists all seven K_i explicitly; completeness is verified by nonzero minors.

The Jung–van der Kulk theorem is then applied: a plane coordinate of degree 3 has the form $P_3(L) + cM$, and one of degree 4 has either the form $P_4(L) + cM$ or $A(M + Q_2(L))^2 + bL + c$, with the appropriate nonzero coefficients. Successive coefficient comparisons in all three projections exclude the possibilities. In the final branch, the nonzero resultant 6561301239 is computed; an explicit Bézout identity with constant 729033471 is also supplied.

The verifier checks 35 exact conditions, and a separate audit repeats the ranks, boundary cases, and parameter eliminations. The structural source of the normal forms is the classical theorem and its degree-divisibility lemma; a convenient primary account is given in [11].

Scope of the assertion. It does not exclude an absolute coordinate of degree 4 whose completion changes all three original coordinates, nor does it exclude relative coordinates of degree ≥ 5.

8 The complete filtration of a one-step transfer

Let $R = k[x]$, $A = R[y,z]$, $(H,W) \in \mathrm{Aut}_R\, R[Y,Z]$, and

$$d = H_Y W_Z - H_Z W_Y \in k^*, \qquad \beta = (p(y) + mz, y), \quad p \in R[y],\ 0 \ne m \in R.$$

Set $h = H \circ \beta$, $w = W \circ \beta$, and $D = h_z \partial_y - h_y \partial_z$. Extract the content of the nonconstant monomials:

$$c = h(x,0,0), \qquad h = c + g h_0,$$

where $g \in R$ is the monic gcd of the coefficients of all monomials of h that are nonconstant in y, z.

Theorem 8.1

We have $g \mid m$ and $B = \ker D = k[x, h_0]$. For every irreducible $\pi \mid m_1 = m/g$, define

$$E_\pi = v_\pi(m_1), \qquad e_\pi = \deg_y(h_0 \bmod \pi) \geq 1.$$

The reduction of h_0 here is independent of z. For $\mathcal{F}_n = \ker D^{n+1}$,

$$\boxed{D^n(\mathcal{F}_n) = \left(g^n \prod_{\pi \mid m_1} \pi^{E_\pi(n - \lfloor n/e_\pi \rfloor)} \right) B.}$$

In particular,

$$\boxed{\operatorname{plinth}(D) = D(A) \cap B = \left(g \prod_{e_\pi \geq 2} \pi^{E_\pi} \right) B.}$$

All multiplicities of the exceptional divisor are retained. Good components with $e_\pi = 1$ disappear from the plinth ideal of the primitive derivation; the factor g is accounted for separately.

Main steps of the proof

1. Kernel and normalization. Over $k(x)$, the pair (h, w) is a coordinate system, and $D(w) = dm$. It follows that $g \mid m$, and the kernel over this field is $k(x)[h_0]$. The reduction of h_0 modulo every prime $\pi \in R$ is nonconstant; hence $R[h_0]$ is π-saturated in A. Successive cancellation of denominators gives $A \cap k(x)[h_0] = R[h_0]$.

2. An exact equation for the modification. Let $(\mathcal{Y}, \mathcal{Z})$ be the inverse of (H, W). The polynomial

$$N(T, V) = \frac{\mathcal{Y}(c + gT, V) - p(\mathcal{Z}(c + gT, V))}{g}$$

does belong to $R[T, V]$, and $\gcd(N, m_1) = 1$; this is proved by reduction modulo g and invertibility of the center over the finite quotient ring. Then

$$A \simeq R[T, V, z]/(m_1 z - N(T, V)), \qquad T = h_0, \quad V = w.$$

Polynomiality of N is essential: formal division without checking it would leave a gap.

3. A monic model at each prime. Localize only R: let $O = R_{(\pi)}$, with coefficient ring $O[T]$. Over $O/(\pi^{E_\pi})$, the center has the form $(G(y), J(y))$ and generates the entire polynomial ring in y. This ring is free of rank e_π over the ring in G; the proof repeats ordinary division and uses a finite lifting process through powers of π. The characteristic polynomial of multiplication by J gives a monic $P_\pi(T, V)$ of degree e_π in V. The equations of the center and P_π generate the same ideal modulo π^{E_π}, leading to the exact model

$$P_\pi(h_0, w) = \pi^{E_\pi} v.$$

Conditions depending on T are retained here: replacing $O[T]$ by its rational function field in T would not suffice.

4. Basis and leading coefficient. The monomials $w^i v^j$, $0 \leq i < e_\pi$, form a basis over $O[h_0]$. Their D-degrees are $i + e_\pi j$ and are pairwise distinct. Thus leading terms cannot cancel. When $n = i + e_\pi j$, we obtain exponent $E_\pi(n - j)$. Intersecting the

local conditions over all primes of R, and using the Chinese remainder theorem to construct preimages, gives the global formula.

The full algebraic proof, including divisibility, monic lifting, and gluing, is provided in `relative_lnd/PROOF.md`; a separate audit is in `literature/independent_image_ideal_review.md`. In addition, 16 examples and 28 separate exact conditions were checked.

An example with a repeated divisor and a nonlinear lift

Let

$$h = y^2 + xy^3 + x^2 z, \qquad D = x^2\partial_y - (2y + 3xy^2)\partial_z.$$

Here $E = 2$ and $e = 2$. For $v = y^4 + xyz - z$,

$$Dv = 2y + xh, \qquad D^2 v = 2x^2, \qquad z = -(1 + xy)v + hy^2.$$

The sequence of image ideals has exponents $0, 2, 2, 4, 4, 6, 6, \ldots$. Finding a single image element yields only an inclusion; equality of the entire ideal follows from the theorem.

Consequences for the theory of LNDs

For $D_{M,e} = x^M\partial_y - ey^{e-1}\partial_z$, $M \geq 1$, $e \geq 2$,

$$I_n(D_{M,e}) = (x^{M(n-\lfloor n/e \rfloor)}).$$

The exponent M is recovered from the multiplicity of the prime divisor of the plinth ideal, and e is recovered as the first n for which $I_n \supsetneq I_1^n$. Thus distinct pairs (M, e) give nonconjugate LNDs. The family itself is classical; no priority is claimed for this formulation of the consequence.

For $h = (x^2 - 2)z - y^3 + 3y$, which arises for $H(4, 5, 7)$,

$$\ker D = k[x, h], \qquad I_n(D) = (x^2 - 2)^{n-\lfloor n/3 \rfloor} k[x, h].$$

This is exact information about this particular LND, not an obstruction to every possible rectifier of the curve.

9 Iterated modifications and Cremona groups

Define

$$r_{-1} = z, \quad r_0 = y, \quad r_j = xr_{j-2} - r_{j-1}^2, \qquad j \geq 1.$$

For a fixed $\ell \geq 1$, set $p = r_\ell$, $w = r_{\ell-1}$, and

$$D_\ell = (-1)^\ell (p_y\partial_z - p_z\partial_y).$$

Then $D_\ell(w) = x^\ell$ and $\ker D_\ell = B = k[x, p]$. The derivation is irreducible: a common divisor of its components must divide x^ℓ, while $p \bmod x = -y^{2^\ell}$ has nonzero derivative with respect to y.

On the open set $x \neq 0$, every operation is invertible; denominators appear when extending across $x = 0$. This gives a concrete connection with Cremona transformations: birational invertibility is straightforward, whereas polynomiality and factorization length are controlled by the exceptional divisor.

Theorem 9.1. The entire filtration

For $I_n(D) = D^n(\ker D^{n+1})$,

$$\boxed{I_n(D_\ell) = x^{a_\ell(n)}B, \qquad a_\ell(n) = \ell n - \sum_{j=1}^{\ell} \left\lfloor \frac{n}{2^j} \right\rfloor .}$$

The proof gives an explicit basis. Set $v_j = r_{\ell-1-j}$ for $0 \le j \le \ell$, and $v_{-1} = p$. Then

$$v_j^2 = xv_{j+1} - v_{j-1} \quad (0 \le j < \ell), \qquad \deg_{D_\ell} v_j = 2^j.$$

By the monic relations, the normal monomials have the form

$$s_n = \prod_{j=0}^{\ell-1} v_j^{\varepsilon_j}\, v_\ell^q, \qquad n = \sum_{j=0}^{\ell-1} \varepsilon_j 2^j + q2^\ell, \quad \varepsilon_j \in \{0,1\}.$$

They form a B-basis and have pairwise distinct D-degrees. When expressed in terms of w, the leading coefficient of v_j is $x^{-(2^j-1)}$. Applying D_ℓ^n and using the binary expansion of n gives the stated sum of floor functions; the factor $n!$ is a unit in k.

For $\ell = 3$, the exponents for $0 \le n \le 8$ are

$$0,\ 3,\ 5,\ 8,\ 9,\ 12,\ 14,\ 17,\ 17.$$

For $\ell \ge 2$, the derivation D_ℓ is not conjugate to any irreducible one-step transfer of Theorem 8.1, even after changing the base coordinate. Indeed, the unique prime divisor of the plinth ideal and its multiplicity ℓ are preserved by every conjugacy. For the one-step class, the exponent of I_2 at this prime would be ℓ when $e = 2$, or 2ℓ when $e > 2$, whereas here it equals $2\ell - 1$.

In particular, D_ℓ is **not triangularizable** for $\ell \ge 2$. If, in some coordinates, it had the form $a(X)\partial_Y + b(X,Y)\partial_Z$ with $a \ne 0$, it would belong to the one-step class with $h = a(X)Z - \int b(X,Y)\,dY$. If the derivative of the first coordinate is a nonzero constant, the plinth ideal equals the entire kernel; when $a = 0$, irreducibility likewise gives a constant translation. Both cases are incompatible with our plinth ideal. The existence of such nontriangularizable examples is not claimed as new; the point here is an explicit way to detect them using two ideals.

The associated graded ring

The same monic relations give a fuller description:

$$\boxed{\operatorname{gr}_{D_\ell} A \simeq B[\xi_0, \dots, \xi_\ell]/(\xi_j^2 - x\xi_{j+1} : j < \ell), \quad \deg \xi_j = 2^j.}$$

After reduction modulo x, the nilradical $N = (\xi_0, \dots, \xi_{\ell-1})$ satisfies $N^\ell \ne 0$, $N^{\ell+1} = 0$, and N/N^2 is free of rank ℓ over $k[p, \xi_\ell]$.

An illustrative example is given by D_3 and xD_2: they have the same plinth ideal (x^3), and the exponents of I_n agree for $n \le 7$. At $n = 8$, they are 17 and 18, respectively. The full filtration detects a difference that the first seven ideals do not.

This illustrates the additional information in the filtration, rather than a newly solved difficult conjugacy problem: the fields are already distinguished by irreducibility. For $\ell = 2$, nontriangularizability is also known from [12, Example 4.19]; the novelty of the general family and formulas is not established here.

Connection with the current question on complexity

In the journal article by Dasgupta–Gaifullin [12, Theorem 4.12], universal level is related to the complexity of a birational representation by $\mathrm{ulev}(D) = c(\alpha) + 1$. Their Question 4.21 concerns a candidate constructed by three alternating modifications and permutations.

Our recurrence allows this type of construction to be studied through its exact filtration. **However, the equality $c(\varphi) = 3$ is not proved here**: it requires excluding every shorter admissible factorization, not merely the canonical one-step class. Extracting such a lower bound from $\mathrm{gr}_D A$ is a substantive next problem.

10 Why relative ML does not distinguish our curves

Let $A = k[x_1, \dots, x_n]$, $I \subsetneq A$, and $\dim(A/I) \le n-2$. For every $r \ge 1$,

$$\boxed{\bigcap_{\substack{0 \neq D \in \mathrm{LND}(A) \\ D(A) \subset I^r}} \ker D = k.}$$

Moreover, the algebra generated by the union of these kernels is A.

For each i, there exists a nonzero $f_i \in I \cap k[x_1, \dots, \widehat{x_i}, \dots, x_n]$: otherwise, $n-1$ coordinate images would be algebraically independent in a ring of dimension $\le n-2$. Take

$$D_i = f_i^r \partial_{x_i}, \qquad \ker D_i = k[x_1, \dots, \widehat{x_i}, \dots, x_n].$$

These LNDs satisfy $D_i(A) \subset I^r$; the intersection of their kernels is k, and their union contains every coordinate. This proves both assertions.

Consequently, the candidate with the weaker condition $D(I) \subset I$ also always gives k for a curve in $\mathbb{A}^3$. Even requiring an arbitrary finite thickening to be fixed pointwise does not help.

For harmonic models, one may take, for example,

$$f_x = C_c(y) - C_b(z), \quad f_y = C_c(x) - C_a(z), \quad f_z = C_b(x) - C_a(y).$$

The mechanism of replicas is consistent with stronger results on flexibility of complements; see Flenner–Kaliman–Zaidenberg [13]. Thus the research strategy is corrected here; no new nontrivial invariant is being claimed.

11 Which named conjectures are affected by rectification

In this section, “refutation” means a complete mathematical argument with exact certificates, presented for verification. External peer review and priority in the literature are not claimed.

Set $K = \mathbb{Q}(\sqrt{-3})$ and

$$\gamma_T(u) = (f(u), g(u), h(u)) = (u^3 - 3u, u^4 - 4u^2, u^5 - 10u).$$

Three equivalent conjectures about a specific object associated with this embedding are formulated in [6, pp. 1319–1320]. They assert, respectively, that the kernel of a

certain LND, the quotient by its slice, and the extended Rees algebra of the curve ideal are not polynomial rings. The last formulation is called Asanuma's Conjecture there. The argument below, applied to the certified rectifying map over K, negates all three formulations over $\mathbb{C}$.

Proposition 11.1 (Explicit coordinates for the associated LND)**.** *Let k be a field of characteristic zero, $F = (P, Q, R_3) \in \mathrm{Aut}_k(\mathbb{A}^3)$, $G = F^{-1}$, $F \circ \gamma = (u, 0, 0)$ and $0 \neq b(t) \in k[t]$. For*

$$A = k[t, u, X_1, X_2, X_3], \qquad D_b = b(t)\partial_u + \sum_i f_i'(u)\partial_{X_i}$$

there is a polynomial coordinate system $A = k[t, p, q, r, s]$ in which $D_b = \partial_s$.

Proof. Set $V_i = f_i(u) - b(t)X_i$ and

$$p = P(V), \qquad q = \frac{Q(V)}{b(t)}, \qquad r = \frac{R_3(V)}{b(t)}, \qquad s = \frac{u - P(V)}{b(t)}.$$

Every numerator is exactly divisible by $b(t)$, since $V \equiv \gamma(u) \pmod{b(t)}$. The inverse substitution is given by

$$u = p + b(t)s, \qquad X_i = \frac{f_i(p + b(t)s) - G_i(p, b(t)q, b(t)r)}{b(t)}.$$

It is also polynomial because $G(p, 0, 0) = \gamma(p)$. Both compositions are the identity after inverting $b(t)$; injectivity of localization proves the same identities in the original ring. Finally, $D_b(V_i) = 0$, so $D_b(p) = D_b(q) = D_b(r) = 0$ and $D_b(s) = 1$. □

For the Shastri model with $b(t) = t$, this gives

$$\ker D = K[t, p, q, r] \simeq K^{[4]}.$$

After extension of scalars, this contradicts the nonpolynomiality asserted in Conjecture 5.1 [6].

Shastri's original parameter polynomial is

$$F_0(x, y, z) = yz - x^3 - 5xy + 2z - 7x,$$

and the slice in Conjecture 5.2 is $\sigma = (u - F_0(V))/t$. Since $D(\sigma) = D(s) = 1$, the difference $\sigma - s$ belongs to $B = \ker D$. Consequently, $A = B[\sigma]$; in particular,

$$A/(\sigma - c) \simeq B \simeq K^{[4]} \qquad (c \in K).$$

This negates Conjecture 5.2 after passing to $\mathbb{C}$. Nevertheless, F_0 itself is not a coordinate of $\mathbb{A}^3$: its gradient vanishes at $(1, -2, 5)$ and $(-1, -2, -5)$. The fact that F_0 is not a coordinate is compatible with σ being a coordinate, since these are different polynomials in different rings.

Finally, with $R_0 = K[x, y, z]$ and $I = I_{\gamma_T}$, we have $R_0 = K[P, Q, R_3]$, $I = (Q, R_3)$, and hence

$$R_0[t, I/t] = K[t, P, Q/t, R_3/t] \simeq K^{[4]}.$$

Here $R_0[t, I/t] \subset R_0[t, t^{-1}]$ is precisely the extended Rees algebra occurring in Conjecture 5.3. This negates the third formulation; it is not a result about arbitrary Rees algebras.

Thus one particular candidate is eliminated in three related presentations. This does not solve the general cancellation problem. Neither does it solve the general rectification problem for $\mathbb{A}^1 \hookrightarrow \mathbb{A}^3$ or the general conjecture that nonrectifiable space lines exist.

Historical distinctions. The conjectured complex nonrectifiability of this specific γ_T is attributed to Shastri in [6, p. 1319]; the supplied rectifying map refutes it. However, Questions Q.1 and Q.2 in [4, p. 12] were already solved by Shastri: they concern polynomial representations of real knots whose complexifications are closed embeddings, rather than their complex rectifiability. Special degree-based conjectures of Abhyankar had earlier refutations, as described in [5, Introduction, pp. 115–116].

Fields and actions. The resulting coordinates are defined over K and hence over $\mathbb{C}$; descent to $\mathbb{Q}$ or $\mathbb{R}$ has not been proved. When $b = t$, the natural action $t \mapsto \lambda t$, $u \mapsto u$, $X_i \mapsto \lambda^{-1} X_i$ is linear in the kernel coordinates, with weights $(1, 0, -1, -1)$. This is a specific linearization. It does not imply statements about all group actions. In the reverse implications toward the embedding, the grading is essential, as in [7]; it cannot be replaced merely by abstract polynomiality of the kernel or by $\mathrm{ML}(B) = k$.

11.1 Full ML, relative ML and the filtration

For a rectified model $B = k[t, p, q, r]$, the four coordinate derivations give

$$\mathrm{ML}(B) = k, \qquad \mathrm{ML}^*(B) = k, \qquad \mathrm{Dk}(B) = B.$$

Here ML^* uses only LNDs with slices, and Dk is the algebra generated by the kernels of all nonzero LNDs. For $D_b = \partial_s$, we have $\mathrm{plinth}(D_b) = B$, $\ker D_b^{n+1} = \bigoplus_{j=0}^{n} B s^j$ and $D_b^n(\ker D_b^{n+1}) = B$. This is consistent with a large minimum slice degree in the original coordinates: these are different characteristics.

For an arbitrary embedding whose rectifying map is unknown, the following dichotomy still holds:

$$\mathrm{ML}(B_b) \in \{k, k[t]\}, \qquad B_b = \ker D_b.$$

Indeed, the derivations ∂_{X_i} commute with D_b and restrict to B_b. The intersection of their kernels there is $B_b \cap k[t, u] = k[t]$. If a nonconstant $a(t)$ belongs to the full ML, then for every LND Δ on B_b, the equality $0 = \Delta a(t) = a'(t)\Delta t$ in a characteristic-zero domain implies $\Delta t = 0$. Thus all of $k[t]$ belongs to ML.

Therefore, $\mathrm{ML}(B_b) = k[t]$ would obstruct rectification. However, $\mathrm{ML}(B_b) = k$ proves neither polynomiality of B_b nor rectifiability. If only LNDs fixing t are allowed from the outset, the relative ML equals $k[t]$ even in the polynomial case. For the new kernels associated with $H(3, 7, 8)$ and $H(4, 5, 7)$, the relevant branch of the dichotomy has not yet been determined.

11.2 What must be retained for the reverse implication

Let $R = k[x, y, z]$, $I = I_\gamma$, $B_\mu = R[t, I/t^\mu]$, $\mu \geq 1$, $\mathrm{wt}(t) = 1$, $\mathrm{wt}(R) = 0$. The embedding is rectifiable if and only if there are homogeneous coordinates

$$B_\mu = k[t, p, q, r], \qquad \mathrm{wt}(t, p, q, r) = (1, 0, -\mu, -\mu)$$

with *the same distinguished element* t. The reverse implication follows from

$$(B_\mu)_0 = R = k[p, t^\mu q, t^\mu r],$$
$$I = t^\mu (B_\mu)_{-\mu} = R \cap tB_\mu = (t^\mu q, t^\mu r).$$

The restriction of p to the curve is an affine parameter and can be normalized to u.

An equivalent formulation is the existence of a homogeneous LND E of degree -1 on B_μ with $E(t) = 1$. The forward implication is supplied by ∂_t. Conversely, the slice theorem and the finite projection

$$\rho(b) = \sum_{j\geq 0} \frac{(-t)^j}{j!} E^j(b)$$

give $B_\mu = (\ker E)[t]$ and a graded isomorphism $\ker E \simeq B_\mu/(t)$. This special fibre is polynomial with weights $(0, -\mu, -\mu)$ by the normal-basis construction in Section 5. Thus the required homogeneous coordinates lift.

For an LND E, without assuming homogeneity, the condition $E(t) = 1$ still gives abstract polynomiality of B_μ, but the preceding proof no longer recovers the embedding. For an LND E satisfying only $E(t) \neq 0$, one obtains only $\mathrm{ML}(B_\mu) = k$. One cannot take an intermediate homogeneous component of an arbitrary LND and declare it locally nilpotent without proof.

Direct connection with the planar theorem. If the ideal I contains a polynomial coordinate, the curve can be placed in a coordinate plane and the Abhyankar–Moh–Suzuki theorem applied. If a parameter polynomial $P \circ \gamma = u$ is itself a coordinate, then with $R = k[P, U, V]$ the required rectifying map is immediately $(P, U - U(\gamma(P)), V - V(\gamma(P)))$. A parameter polynomial that is not necessarily a coordinate is insufficient for either argument.

12 A normal Euler field: a complete constructive criterion

Let $R = k[Y_1, \dots, Y_n]$, let $\gamma : \mathbb{A}^1 \hookrightarrow \mathbb{A}^n$ be a closed embedding, and put $I = I_\gamma$, $B = R[t, I/t]$. For a derivation δ, consider the conditions

$$\delta(R) \subset I, \qquad (\delta - \mathrm{id})(I) \subset I^2. \tag{1}$$

They say that the field vanishes on the curve and acts as the identity on first-order normal functions. Then

$$E = \partial_t + t^{-1}\delta$$

preserves B, since $E(Y_i) = \delta Y_i/t \in B$ and $E(F/t) = (\delta F - F)/t^2 \in B$ for $F \in I$. Local finiteness of δ means that $\mathrm{span}_k\{\delta^j F : j \geq 0\}$ is finite-dimensional for every $F \in R$.

Theorem 12.1. *For a fixed δ satisfying (1), the following are equivalent:*

1. *E is an LND on B;*
2. *δ is locally finite on R;*
3. *there are coordinates $R = k[p, q_1, \dots, q_{n-1}]$ such that $p\gamma = u$, $I = (q_1, \dots, q_{n-1})$, and $\delta = \sum_j q_j \partial_{q_j}$.*

Consequently, the existence of such a locally finite field is equivalent to rectifiability of γ over k.

Proof. In the localization, direct induction gives

$$E^m F = t^{-m} \prod_{j=0}^{m-1} (\delta - j)F, \qquad E^m(F/t) = t^{-m-1} \prod_{j=1}^{m} (\delta - j)F. \tag{2}$$

If E is locally nilpotent, every F is annihilated by a nonzero polynomial in δ, which gives local finiteness.

The Leibniz rule and (1) give

$$(\delta - m)I^m \subset I^{m+1} \quad (m \geq 0).$$

Now suppose that δ is locally finite. Let V be a finite-dimensional δ-stable space, and let M be the largest ordinary degree of its elements. Since $I \subset \mathfrak{m}_{\gamma(0)}$, we have $V \cap I^{M+1} = 0$. Thus $\delta(\delta-1)\cdots(\delta-M)$ annihilates V. Its roots are distinct and belong to k: semisimplicity and nonnegative integer weights follow automatically, without passage to the algebraic closure. We obtain

$$R = \bigoplus_{d \geq 0} R_d, \qquad \delta|_{R_d} = d.$$

For a nonzero eigenvector, its exact I-adic order equals its weight, because the induced action on I^m/I^{m+1} is multiplication by m. The ideals I^m are graded: the spectral projectors are polynomials in δ. Hence

$$I^m = \bigoplus_{d \geq m} R_d, \quad R_0 \simeq R/I \simeq k[u], \quad R_1 \simeq I/I^2, \quad R = R_0[R_1].$$

The last equality follows by induction on the weight: for $d \geq 2$ we have $R_d \subset I^2$, and each homogeneous factor in a product has a smaller positive weight.

Freeness of I/I^2 does not require homological algebra. Choose $P\gamma = u$, and put $v = \gamma'$, $a = \nabla P(\gamma)$. Then $av = 1$, and the row module $L = \{\ell \in k[u]^n : \ell v = 0\}$ is free of rank $n-1$ over the PID $k[u]$. The gradient defines an isomorphism

$$I/I^2 \longrightarrow L, \qquad F \longmapsto \nabla F(\gamma).$$

An explicit lift of a row is

$$F_\ell(Y) = \sum_i \ell_i(P(Y))(Y_i - f_i(P(Y))).$$

Injectivity follows from Taylor expansion: a function whose value and first gradient both vanish on the curve belongs to I^2. Choose an invariant lift p of the parameter u and a basis q_j of R_1 over $R_0 = k[p]$. Then $R = k[p, q_1, \dots, q_{n-1}]$; algebraic independence follows from transcendence degree n. This gives the normal form of δ. Finally, $B = k[t, p, q_1/t, \dots, q_{n-1}/t]$, and $E = \partial_t$ in these coordinates. This proves all the implications. □

12.1 Finite verification and reconstruction of the automorphism

It suffices to exhibit the exact stability condition

$$V_N = \operatorname{span}_k\{1, \delta^j Y_i : 0 \leq j \leq N, 1 \leq i \leq n\}, \qquad \delta^{N+1} Y_i \in V_N.$$

Spaces spanned by products of elements of V_N are stable by the Leibniz rule, so δ is locally finite on all of R. If the degrees of elements of V_N are at most M, the spectral projection of a polynomial F onto weight j is computed by

$$\operatorname{pr}_j(F) = \prod_{\substack{0 \leq i \leq L \\ i \neq j}} \frac{\delta - i}{j - i} F, \qquad L \geq M \deg F.$$

Thus $p = \mathrm{pr}_0(P)$, $q_j = \mathrm{pr}_1(F_{\ell_j})$ are the required coordinates. The inverse map is reconstructed from the weight components of Y_i and finitely many transverse derivatives. The detailed formula is retained in `euler/theory.md`. No universal a priori bound on N is given here.

A positive control reconstructs the nonlinear coordinates

$$r = z - x^2, \qquad q = y - r^2, \qquad p = x - q^2.$$

The corresponding space V_4 has dimension 9 and is exactly stable. This checks the implementation; it does not replace the general proof.

12.2 Necessary conditions that are still insufficient

The normal form implies

$$\operatorname{div} \delta = n - 1, \qquad (\delta Y_1, \dots, \delta Y_n) = I.$$

The equality of ideals describes the entire fixed-point scheme, rather than only its set of points. Constancy of the divergence of an LF field is a classical fact [14]; no priority for this fact is claimed.

Even both conditions together with (1) are insufficient. For the line $I = (y, z)$, the field

$$\delta = yz\partial_x + y\partial_y + (z + y^2x)\partial_z$$

passes all these tests. However, $\delta^{2m}x$ contains xy^{3m} with a positive coefficient. It arises by repeating the path $x \mapsto yz \mapsto y^3x$; all coefficients are nonnegative, so no cancellation occurs. The degrees are unbounded, hence the field is not LF. Local finiteness of the normal field must not be confused with local nilpotence: in the correct normal form $\delta^m q_j = q_j$, while E is the locally nilpotent derivation.

13 The exact limitation of the naive normal field

For any parameter polynomial $P\gamma = u$, put

$$\delta_P = \sum_i (Y_i - f_i(P))\partial_{Y_i}.$$

This field always satisfies (1), but has the following essential limitation.

Theorem 13.1. *There exists $P\gamma = u$ for which δ_P is locally finite if and only if there exists an affine-linear L such that $L\gamma = u$.*

Proof. If $P = L$ is affine-linear, then $\delta_P(P) = 0$ and $\delta_P^2 Y_i = \delta_P Y_i$, giving local finiteness on the generators. Conversely, the necessary divergence identity is

$$\sum_i f_i'(P)\partial_i P = 1.$$

By an affine change of coordinates, make the positive degrees of the nonconstant f_i pairwise distinct; the remaining components can be made zero. The form of δ_P is preserved under this change. If there is no linear parameter, all positive degrees are at least 2. Let $d = \deg P$, and let j correspond to the largest degree of an f_j with $\partial_j P \neq 0$. The degree of this summand is at least $(\deg f_j - 1)d$, whereas each preceding

summand has degree at most $(\deg f_i)d-1 \le (\deg f_j-1)d-1$. The leading term cannot cancel, so the sum cannot equal 1. If there are no active nonzero components, the sum is zero, which is also impossible. □

For harmonic models with distinct indices $a, b, c \ge 2$, there is no linear parameter. Thus *no* choice of P, of any degree, makes this field LF. The exclusion also applies to the already rectified $H(3,4,5)$ and $H(3,5,7)$, so it is not an obstruction to rectifiability of the curve itself.

14 General normal fields: exact spaces of candidates

We seek all $\delta = a_1\partial_x + a_2\partial_y + a_3\partial_z$ with $\deg a_i \le d$, satisfying (1) and $\operatorname{div}\delta = 2$. Each a_i is expanded in a full basis of $I \cap R_{\le d}$. Writing $v = \gamma'$, the second normal condition is equivalent to the columns of $J_a(\gamma)-\mathrm{Id}$ being multiples of the unimodular v. This gives a rational linear system $Ac = b$ covering all fields of the specified degree bound.

Model	d	$\dim I_{\le d}$	Unknowns	$\operatorname{rank} A$	Result
$H(3,7,8)$	7	63	189	189	inconsistent
$H(3,7,8)$	8	100	300	276	family 24
$H(4,5,7)$	6	42	126	—	inconsistent
$H(4,5,7)$	7	71	213	194	family 19

Inconsistency is certified by an exact rational vector satisfying $\lambda^{\mathsf{T}}A = 0$, $\lambda^{\mathsf{T}}b = 1$; the conclusion therefore persists under any characteristic-zero field extension. For the consistent systems, a rational point and a full basis of directions are provided. Exact null vectors give upper bounds on rank; matching modular lower bounds certify completeness.

Thus the first degrees at which the *normal conditions together with global divergence* 2 are consistent are 8 and 7. These are not minimum degrees of LF fields or rectifiers. For $H(4,5,7)$, degree 6 is already impossible without the divergence condition. For $H(3,7,8)$, the normal conditions alone at degree 7 have an 11-dimensional family; it is global divergence that excludes it.

14.1 Two selected representatives are rigorously excluded

In the $H(3,7,8)$ family, the selected rational representative preserves (x,y) and acts on the quotient $k[z]$ as $\psi(z)\partial_z$, with $\deg\psi = 6$. In the $H(4,5,7)$ family, the selected representative preserves (y,z) and induces $\varphi(x)\partial_x$, with $\deg\varphi = 7$. For a polynomial a of degree $e \ge 2$ in characteristic zero,

$$\deg((a\partial_v)^m v) = 1 + (e-1)m, \qquad \operatorname{lc}((a\partial_v)^m v) = \operatorname{lc}(a)^m \prod_{j=0}^{m-1}(1+(e-1)j) \neq 0.$$

Hence neither representative is LF, and the corresponding E are not LNDs. The full coefficients and separate checks are included in the JSON certificates. The axes are not preserved by the entire families: respectively 13 of 24 and 11 of 19 selected basis directions fail to preserve them. These counts concern the chosen bases; they are not dimensions of excluded subfamilies. No exclusion of all complex parameter values is established.

14.2 Real topology and conjugation

By [10, p. 2, Fig. 3], $H(3,7,8)$ has type $T(2,5)$ up to mirror image, whereas $H(4,5,7)$ has type 5_2. Consequently, these models admit no real polynomial rectifiers. By Theorem 12.1, *every* suitable LF field for them must have nonreal coefficients. This is a known topological restriction on the field of definition, not a complex obstruction.

If a suitable complex δ commuted with $\bar{\delta}$, their average would be a real LF field satisfying the same normal conditions. Commuting LF operators have finite-dimensional joint cyclic spaces, so the average is indeed LF. This would give the forbidden real rectification. Similarly, commuting E and $\bar{E}$ would give a real LND with slice t. Therefore, both commutators are nonzero in these cases. Moreover,

$$[E,\bar{E}] = t^{-2}([\delta,\bar{\delta}] + \delta - \bar{\delta}),$$

so the two commutation conditions must not be identified.

15 What remains hypothetical and how to proceed

The results established here resolve specific questions for specific models and classes of derivations. The following table records questions that must not be listed as solved.

Problem	Precise status and next verifiable step
All harmonic embeddings	Two infinite families and finite checks are provided. A general passage between residue classes or a new constructive rectification is needed. A bounded search is not a classification.
Complex rectification of $H(3,7,8)$ and $H(4,5,7)$	Not constructed. One can search for a locally finite field in the affine spaces of candidates obtained here. Absence of such a field at these degrees, if proved, would not exclude fields of higher degree.
Full ML of the new kernels	At present there is a dichotomy: k or $k[t]$. For the value k, one LND moving t suffices. For the value $k[t]$, one must prove that every LND fixes t; a bounded search cannot prove this.
Homogeneous E with slice t	The conditions $E(t) = 1$ and preservation of the kernel do not yet imply local nilpotence. Vanishing of iterates on generators is an exact positive certificate; failure of finitely many iterates to vanish does not prove that no suitable field exists.
Formal coordinates	All finite thickenings and the completion have been constructed. Termination of the coordinates as polynomials has not been proved: this is a separate global problem.
Dasgupta–Gaifullin Question 4.21	Filtration formulas have been obtained, but minimality of a three-step birational presentation has not been established. All admissible shorter presentations must be excluded.

Problem	Precise status and next verifiable step
Minimum degrees of rectifiers	The degrees of the exhibited automorphisms are upper bounds. Minimum slice degrees do not replace this problem.
Real descent	Real rectification of the nontrivial knots under consideration is impossible. This alone does not exclude polynomiality of the corresponding real kernel without preservation of the distinguished t.
General cancellation and linearization problems	One proposed candidate has been ruled out; the general problems and all possible actions have not been fully investigated.

Why the search for an LND with a slice is motivated

In B_μ, the element t is distinguished by the embedding itself, while the special fiber $t = 0$ already has a polynomial description. If one can exhibit a homogeneous derivation E of degree -1 with $E(t) = 1$ and prove its local nilpotence, the slice formula gives

$$\rho_E(f) = \sum_{j\geq 0} \frac{(-t)^j}{j!} E^j(f), \qquad B_\mu = (\ker E)[t].$$

The sum is finite precisely because of local nilpotence. Under homogeneity, this construction recovers coordinates for the original embedding, as proved in §11. Thus the additional LND requirement serves a precise geometric purpose. It is not a technical detail that follows automatically from the linear equations on coefficients.

For a finite generating set $b_1, \dots, b_m$, it suffices to exhibit integers N_i and identities $E^{N_i}(b_i) = 0$. The Leibniz rule then gives local nilpotence on the whole algebra. For the search for a normal locally finite field, an alternative positive certificate is a finite-dimensional stable space containing the generators, as in §12. Both methods yield a verifiable proof when successful, but no general time or degree bound for these searches is proved here.

The role of Cremona groups

Birational changes of the form

$$(x, y, z) \longmapsto (x, y, m(x)z - q(x, y))$$

provide an effective language for transferring fields and coordinates after localization. Returning to the polynomial ring requires exact divisibility and a check of the modification center. The obstruction established for the natural center of the family $H(3, 3n + 1, 3n + 2)$ shows that merely having a rational inverse is insufficient. The practical programme consists of searching for other centers, computing image ideals and comparing different factorizations. No general classification of Cremona subgroups is claimed here.

Connection with the Abhyankar–Moh theorem

A direct application of the planar theorem becomes possible once a coordinate surface containing the curve has been exhibited: on this surface, the embedded line

becomes a planar embedding to which [2, 3] applies. Finding such a surface in the ambient space is the substantive missing step. The equivalent criteria through a coordinate parameter, a graded LND and a normal locally finite field offer different methods for addressing precisely this step. It must not be replaced by an appeal to the already known planar theorem.

Suggested reading sequence

For an introduction to LNDs, a useful starting point is *First Principles* and *Further Properties of LNDs* in Freudenburg's book [15], followed by the chapters *Dimension Three, Algorithms, Makar-Limanov and Derksen Invariants* and *Slices, Embeddings and Cancellation*. For this project, the next readings are [6] (the precise conjectures and the kernel ring), [7] (equivariant structure), [8] (relative coordinates), [12] (iterations of LNDs and complexity), and [9, 10] (harmonic curves and knots).

16 Reproducibility and contents of the supplement

All computational files are in `computations/`. Comments in the supplied code are in English. The original versions are retained in the provenance archive; a separate check verifies that edits to Python comments and documentation strings do not change the abstract syntax tree of the executable part. This is a useful packaging check, but not a proof of algorithmic correctness.

Directory	Purpose
`baseline/`	Original rectifiers; exact checks of mutually inverse maps, printed components and original certificates.
`legacy_appendix/`	Computations from the published appendix of 26 October, including local dependencies for offline execution.
`continuation/`	Fourteen continuation checks: minimum slice degrees, multiplicity-dependent degrees, formal coordinates, transfer restrictions, LND filtrations and relative ML.
`euler/`	Normal fields, ranks of linear systems, dual inconsistency certificates and invariant-axis obstructions for selected fields.
`audit_checks/`	Verification identifying the candidate in Conjectures 5.1–5.3 with the original model and original slice.
`verification_logs/`	Standard output, errors and execution time for every program in the repeated run.
`provenance/`	Original versions, checksums and information on the origin of the materials.

The entry point is `computations/run_all_checks.py`. Run the following from the `computations/` directory:

```
python -m pip install -r requirements.txt
python run_all_checks.py
```

The fresh run of 28 October 2026 completed successfully: **35 out of 35 programs**. All principal algebraic checks are exact, without replacing identities by numerical approximations. The final machine-readable record `verification_summary.json` records the results and commands. The number of programs is not the number of theorems: one program may check many identities, while infinite families require the mathematical proofs given above.

GAP/Macaulay2 code and historical logs are included for independent checking, but those systems were not run in the current environment. Their older successful logs are explicitly distinguished from the new Python run. Lengthy search experiments are retained as source code and data; they do not replace the shorter verification programs.

Compatibility with the repository

The baseline was checked at commit `b954f3609e900761fddc9c8ff03d8301e9604cab`. All 75 files of the published appendix matched the Git objects of that version. The update adds only a separate directory, `updates/2026-10-28/`, and a pointer to it within the earlier appendix. The main preprint and earlier certificates are not replaced.

Two earlier research directions are clarified substantively. First, the relative ML invariant in the earlier appendix was proposed conditionally; §10 shows that the anticipated nontriviality is impossible under that definition. Second, the new degree bound ≥ 5 for $H(4,5,7)$ concerns coordinates preserving one of the original variables. The general search for an absolute coordinate starting at degree 4 remains justified. This clarification does not require editing the archived text, but it should accompany future citations of that text.

Verification by the reader

First check the precise model and its coefficient field; then run the rectifier certificates; next check the constructive passage to the kernel and slice. For new general statements, the principal object to verify remains the mathematical proof. If an error is found, please identify the model, the statement, the execution command and the full log. The authors invite readers to undertake this verification and to develop the research programme.

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